

Plasma antioxidant capacity changes following a meal as a measure of the ability of a food to alter in vivo antioxidant status.

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OBJECTIVE: Determine 1) if consumption of a meal of different fruits or berries increases plasma hydrophilic (H-) or lipophilic (L-) antioxidant capacity (AOC) measured as Oxygen Radical Absorbance Capacity (ORAC(FL)); 2) if including macronutrients in the meal alters postprandial changes in AOC; and 3) if preliminary recommendations can be developed for antioxidant intake. METHODS: Changes in plasma AOC following consumption of a single meal of berries/fruits (blueberry, dried plum, dried plum juice, grape, cherry, kiwifruit and strawberry) were studied in 5 clinical trials with 6-10 subjects per experiment. In two studies with blueberry or grape, additional macronutrients (carbohydrate, fat, protein) were included in the control and treatment meals. Blood samples collected before and after the meal were analyzed for AOC. RESULTS: Consumption of dried plums or dried plum juice did not alter either the H- or L-AOC area under the curve (AUC). Consumption of blueberry in 2 studies and of mixed grape powder [12.5 (Study #1), 39.9 (Study #4) and 8.6 (Study #5) mmole Trolox Equivalents (TE) AOC, respectively] increased hydrophilic AOC AUC. L-AOC increased following a meal of blueberry containing 12.5 mmole TE AOC (Study #1). Consumption of 280 g of cherries (4.5 mmol TE AOC) increased plasma L-AOC but not H-AOC. The AOC in the control groups in which additional macronutrients (Studies #4 and #5) were added decreased from the postprandial baseline AOC measurement. CONCLUSION: We have demonstrated that consumption of certain berries and fruits such as blueberries, mixed grape and kiwifruit, was associated with increased plasma AOC in the postprandial state and consumption of an energy source of macronutrients containing no antioxidants was associated with a decline in plasma AOC. However, without further long term clinical studies, one cannot necessarily translate increased plasma AOC into a potential decreased risk of chronic degenerative disease. Preliminary estimates of antioxidant needs based upon energy intake were developed. Consumption of high antioxidant foods with each meal is recommended in order to prevent periods of postprandial oxidative stress.